

STAT 153 & 248 - Time Series

Lecture Three

Fall 2026, UC Berkeley

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03 September 2026

We shall look at the math behind the linear regression calculations. We will mostly focus on the Bayesian approach which is conceptually simpler and generalizes easily to more complicated models. We shall look at the main frequentist ideas in the next lecture.

In order to motivate the main ideas behind Bayesian inference for linear regression, we shall start with some simple inference problems first before looking at linear regression.

1 Simple Question 1

Suppose a and b are two numbers and all we know is that $a+b = 7$. Based on this information, what can we say about a ?

There is clearly not enough information to determine a . For every choice of a , we can choose $b = 7 - a$. Thus there are infinitely many possible pairs (a, b) .

2 Simple Question 2

Suppose a and b are two numbers such that $a + b = 7$. Suppose b is small in magnitude. What then can be said about a ?

Here is a way of arriving at a reasonable solution to this problem using probability theory. Assume that a and b are independent with

$$a \sim \text{Unif}(-C, C) \quad \text{and} \quad b \sim N(0, \sigma^2) \tag{1}$$

where C is very large, and σ is a fixed quantity that characterizes smallness of b . Depending on what we understand by smallness of b , we can set a value of σ (e.g., $\sigma = 0.5$).

Using the *modeling* assumption (1) and the available *data* that $a + b = 7$, our uncertainty in a will be captured by the probability distribution:

$$a \mid a + b = 7.$$

Here is a simple way of calculating this conditional distribution. Let $y = a + b$ so that we need to calculate the conditional density of $a \mid y = 7$. The conditional distribution of $y \mid a$

is given by $N(a, \sigma^2)$ because $y = a + b$, and b is distributed as $N(0, \sigma^2)$ independently of a . Therefore by Bayes rule:

$$\begin{aligned} f_{a|y=7}(a) &\propto f_{y|a}(7) f_a(a) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(7-a)^2}{2\sigma^2}\right) \times \frac{I\{-C < a < C\}}{2C} \\ &\propto \exp\left(-\frac{(7-a)^2}{2\sigma^2}\right) I\{-C < a < C\}. \end{aligned}$$

This is the normal distribution $N(7, \sigma^2)$ truncated to the interval $(-C, C)$. Because we shall take C to be very large, the restriction to $(-C, C)$ has almost no effect and therefore,

$$a \mid a + b = 7 \approx N(7, \sigma^2).$$

Thus under our probabilistic assumptions and the given information, our uncertainty about a is captured by the normal distribution centered at 7 and variance σ^2 . The amount of uncertainty is controlled by σ^2 .

3 Simple Question 3

Now suppose we have three numbers a, b_1, b_2 . Assume that both b_1 and b_2 are small in magnitude. Also suppose that

$$a + b_1 = 7 \quad \text{and} \quad a + b_2 = 10.$$

What can then be said about a ?

Note now that there are three unknown numbers and two pieces of information. So this problem is also logically unanswerable without further assumptions. We shall make probabilistic assumptions on the unknown a, b_1, b_2 . Specifically, we assume that

$$a \sim \text{Unif}(-C, C), \quad \text{and} \quad b_1, b_2 \mid \sigma \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \quad \text{and} \quad \log \sigma \sim \text{Unif}(-C, C) \quad (2)$$

where, as before, C is a very large constant. The idea behind this modeling assumption is:

1. We don't want to assume anything specific about a
2. Because b_1, b_2 are given to be small in magnitude, we assume a normal distribution centered at 0 for them with their magnitude controlled by a parameter σ . This also assumes they have the same scale.
3. The common scale σ of b_1, b_2 is also unknown. We don't want to assume anything specific about this scale so we use the uniform $(-C, C)$ prior on $\log \sigma$. Because σ is always positive, $\log \sigma$ makes sense and it takes values that can be both positive and negative, and this makes the $\text{Unif}(-C, C)$ prior reasonable for $\log \sigma$.

The assumption $\log \sigma \sim \text{unif}(-C, C)$ leads to the following density for σ (by the change of variables formula): The last assumption says that we do not know the order of magnitude of σ . By a change of variables, it gives the density

$$f_\sigma(\sigma) = f_{\log \sigma}(\log \sigma) \left| \frac{d}{d\sigma} \log \sigma \right| = f_{\log \sigma}(\log \sigma) \frac{1}{\sigma} = \frac{I\{-C < \log \sigma < C\}}{2C\sigma}.$$

With these assumptions, we need to calculate the conditional distribution of:

$$a \mid a + b_1 = 7, a + b_2 = 10.$$

This determines our uncertainty about a given the information $a + b_1 = 7$ and $a + b_2 = 10$. To compute the above conditional, let us use the notation:

$$y_1 = a + b_1 \quad \text{and} \quad y_2 = a + b_2,$$

so that we need to calculate the density of:

$$a \mid y_1 = 7, y_2 = 10. \quad (3)$$

One complication in calculating this density is the variable σ which is present in the modeling assumption but is not present in this conditional density. A simple way of dealing with this is to first compute the joint conditional density

$$a, \sigma \mid y_1 = 7, y_2 = 10 \quad (4)$$

and then to integrate over σ to obtain the required conditional density (3). To calculate (4), we use the Bayes rule in the following way: (note that $y_1, y_2 \mid a, \sigma \sim N(a, \sigma^2)$)

$$\begin{aligned} f_{a, \sigma \mid y_1, y_2}(a, \sigma) &\propto f_{y_1, y_2 \mid a, \sigma}(y_1, y_2) f_{a, \sigma}(a, \sigma) \\ &= \left\{ \prod_{j=1}^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_j - a)^2}{2\sigma^2}\right) \right\} f_a(a) f_\sigma(\sigma) \\ &= \left\{ \prod_{j=1}^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_j - a)^2}{2\sigma^2}\right) \right\} \frac{I\{-C < a < C\}}{2C} \frac{I\{-C < \log \sigma < C\}}{2C\sigma} \\ &\propto \sigma^{-3} \exp\left(-\frac{(y_1 - a)^2 + (y_2 - a)^2}{2\sigma^2}\right) I\{-C < a < C\} I\{e^{-C} < \sigma < e^C\} \end{aligned}$$

where, in the last step above, we ignored the terms involving $\sqrt{2\pi}$ and C in the proportional constant. As C is very large, we shall also ignore the first indicator function above because it will be 1 for all reasonable values of a , and replace the second indicator function by $I\{\sigma > 0\}$ because it will be 1 for all reasonable values of $\sigma > 0$. This gives

$$f_{a, \sigma \mid y_1, y_2}(a, \sigma) \propto \sigma^{-3} \exp\left(-\frac{(y_1 - a)^2 + (y_2 - a)^2}{2\sigma^2}\right) I\{\sigma > 0\}.$$

This conditional density reflects our knowledge of both a and σ given the information on y_1 and y_2 . Since the question only cares about a (and not σ), we can obtain (3) by integrating the above with respect to σ . This gives:

$$f_{a \mid y_1, y_2}(a) = \int f_{a, \sigma \mid y_1, y_2}(a, \sigma) d\sigma \propto \int_0^\infty \sigma^{-3} \exp\left(-\frac{(y_1 - a)^2 + (y_2 - a)^2}{2\sigma^2}\right) d\sigma.$$

Let

$$S(a) = (y_1 - a)^2 + (y_2 - a)^2,$$

and we can call $S(a)$ the sum of squares function. Then

$$f_{a \mid y_1, y_2}(a) \propto \int_0^\infty \sigma^{-3} \exp\left(-\frac{S(a)}{2\sigma^2}\right) d\sigma.$$

We use the change of variable:

$$\sigma = t\sqrt{S(a)} \quad \text{so that} \quad d\sigma = \sqrt{S(a)} dt. \quad (5)$$

Then

$$\begin{aligned} f_{a|y_1, y_2}(a) &\propto \int_0^\infty \sigma^{-3} \exp\left(-\frac{S(a)}{2\sigma^2}\right) d\sigma \\ &= \int_0^\infty t^{-3} S(a)^{-3/2} \exp\left(-\frac{1}{2t^2}\right) \sqrt{S(a)} dt \\ &= \frac{1}{S(a)} \int_0^\infty t^{-3} \exp\left(-\frac{1}{2t^2}\right) dt \\ &\propto \frac{1}{S(a)}, \end{aligned}$$

where, in the last step, we ignored the integral term because it is a constant (i.e., does not depend on a). We therefore proved

$$f_{a|y_1, y_2}(a) \propto \frac{1}{(y_1 - a)^2 + (y_2 - a)^2}.$$

It turns out that this is a Cauchy density. To see this, first complete the square and rewrite $S(a)$ as:

$$(y_1 - a)^2 + (y_2 - a)^2 = 2 \left(a - \frac{y_1 + y_2}{2} \right)^2 + \frac{(y_1 - y_2)^2}{2} = 2 \left\{ \left(a - \frac{y_1 + y_2}{2} \right)^2 + \left(\frac{|y_1 - y_2|}{2} \right)^2 \right\}$$

so that

$$f_{a|y_1, y_2}(a) \propto \frac{1}{\left(a - \frac{y_1 + y_2}{2} \right)^2 + \left(\frac{|y_1 - y_2|}{2} \right)^2}. \quad (6)$$

The density of a Cauchy random variable with location μ and scale τ is

$$\text{Density of Cauchy}(\mu, \tau) = \frac{1}{\pi\tau} \frac{1}{1 + ((x - \mu)/\tau)^2} \propto \frac{1}{\tau^2 + (x - \mu)^2}. \quad (7)$$

Comparing (6) and (7), we get

$$a \mid y_1, y_2 \sim \text{Cauchy} \left(\frac{y_1 + y_2}{2}, \frac{|y_1 - y_2|}{2} \right).$$

For $y_1 = 7$ and $y_2 = 10$, this becomes

$$a \mid y_1 = 7, y_2 = 10 \sim \text{Cauchy}(8.5, 1.5).$$

It is quite remarkable that we obtain a Cauchy density to capture the uncertainty about a given the observations. This is basically a consequence of the modeling assumptions (2).

4 Question 4: Several Measurements of One Quantity

Suppose a scientist makes $n = 6$ measurements of an unknown physical quantity θ and obtains

$$y_1 = 26.6, \quad y_2 = 38.5, \quad y_3 = 34.4, \quad y_4 = 34.0, \quad y_5 = 31.0, \quad y_6 = 23.6.$$

Based on these observations, what can be inferred about θ ?

We consider each observation y_i as comprised of θ and an 'error' or 'noise' term ϵ_i :

$$y_i = \theta + \epsilon_i \quad \text{for } i = 1, \dots, n.$$

This means that we are trying to write the n observations $y_1, \dots, y_n$ in terms of $n + 1$ unknowns θ and $\epsilon_1, \dots, \epsilon_n$. On these $n + 1$ unknowns, we make the following modeling assumptions:

$$\theta \sim \text{Unif}(-C, C) \quad \text{and} \quad \epsilon_1, \dots, \epsilon_n \mid \sigma \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \quad \text{and} \quad \log \sigma \sim \text{Unif}(-C, C).$$

This assumption is the same as that used in the previous section, the only difference is that n is now 6 as opposed to 2, and also we are using different notation (θ in place of a , and $\epsilon_1, \dots, \epsilon_n$ in place of b_1, b_2).

Our goal is to find the conditional density of θ given $y_1, \dots, y_n$. As in question 3, we first find the joint conditional density of (θ, σ) given $y_1, \dots, y_n$, and then we integrate out σ to calculate $\theta \mid y_1, \dots, y_n$. By Bayes rule,

$$\begin{aligned} f_{\theta, \sigma \mid y_1, \dots, y_n}(\theta, \sigma) &\propto f_{y_1, \dots, y_n \mid \theta, \sigma}(y_1, \dots, y_n) f_{\theta, \sigma}(\theta, \sigma) \\ &= \left\{ \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_j - \theta)^2}{2\sigma^2}\right) \right\} f_{\theta}(\theta) f_{\sigma}(\sigma) \\ &= \left\{ \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_j - \theta)^2}{2\sigma^2}\right) \right\} \frac{I\{-C < \theta < C\}}{2C} \frac{I\{-C < \log \sigma < C\}}{2C\sigma} \\ &\propto \sigma^{-n-1} \exp\left(-\frac{(y_1 - \theta)^2 + \dots + (y_n - \theta)^2}{2\sigma^2}\right) I\{-C < \theta < C\} I\{e^{-C} < \sigma < e^C\}. \end{aligned}$$

Ignoring the indicators, we get

$$f_{\theta, \sigma \mid y_1, \dots, y_n}(\theta, \sigma) \propto \sigma^{-n-1} \exp\left(-\frac{(y_1 - \theta)^2 + \dots + (y_n - \theta)^2}{2\sigma^2}\right) I\{\sigma > 0\}.$$

Integrating the above with respect to σ , we get

$$f_{\theta \mid y_1, \dots, y_n}(\theta) = \int f_{\theta, \sigma \mid y_1, \dots, y_n}(\theta, \sigma) d\sigma \propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{(y_1 - \theta)^2 + \dots + (y_n - \theta)^2}{2\sigma^2}\right) d\sigma.$$

Let

$$S(\theta) = (y_1 - \theta)^2 + \dots + (y_n - \theta)^2$$

be the sum of squares function. Then

$$f_{\theta \mid y_1, \dots, y_n}(\theta) \propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\theta)}{2\sigma^2}\right) d\sigma.$$

Using the same change of variable as in (5), we deduce

$$f_{\theta \mid y_1, \dots, y_n}(\theta) \propto \left(\frac{1}{S(\theta)}\right)^{n/2}. \quad (8)$$

We will see next week that this is a t -density with $n - 1$ degrees of freedom. Note that the value of θ that minimizes $S(\theta)$ is the sample mean:

$$\bar{y} = \frac{y_1 + \dots + y_n}{n}.$$

Because $1/S(\theta)$ appears in (8), it means that the conditional density of θ given $y_1, \dots, y_n$ is maximized at the sample mean.

5 Linear Regression with Time as a Covariate

Suppose $y_1, \dots, y_n$ is a time series. We want to decompose this time series as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where x_i is known. The simplest example is $x_i = i$; this corresponds to regression with time as covariate. The analysis also works when $x_i = y_{i-1}$ as in lagged regression but we shall study this later. There are $n + 2$ unknowns here: β_0, β_1 and $\epsilon_1, \dots, \epsilon_n$. The goal is to figure them out from the n known observations $y_1, \dots, y_n$.

We make the following modeling assumptions:

$$\beta_0 \sim \text{Unif}(-C, C) \quad \beta_1 \sim \text{Unif}(-C, C) \quad \epsilon_1, \dots, \epsilon_n \mid \sigma \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2) \quad \log \sigma \sim \text{Unif}(-C, C).$$

Our interest is usually in β_0, β_1 . So we aim to calculate the conditional density of (β_0, β_1) given $y_1, \dots, y_n$. We do this by first calculating the density of $(\beta_0, \beta_1, \sigma)$ given $y_1, \dots, y_n$, and then integrating out σ . By Bayes rule,

$$\begin{aligned} f_{\beta_0, \beta_1, \sigma \mid y_1, \dots, y_n}(\beta_0, \beta_1, \sigma) &\propto \left\{ \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right] \right\} \frac{I\{-C < \beta_0 < C\}}{2C} \frac{I\{-C < \beta_1 < C\}}{2C} \\ &\quad \times \frac{I\{-C < \log \sigma < C\}}{2C\sigma} \\ &\propto \sigma^{-(n+1)} \exp \left\{ -\frac{S(\beta_0, \beta_1)}{2\sigma^2} \right\} I\{\sigma > 0\}, \end{aligned}$$

where

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

is the sum of squares, and we have ignored the indicator $I\{-C < \beta_0 < C\}$, $I\{-C < \beta_1 < C\}$ and replaced the indicator $I\{e^{-C} < \sigma < e^C\}$ by $I\{\sigma > 0\}$.

As in the previous two sections, integrating out σ gives

$$f_{\beta_0, \beta_1 \mid y_1, \dots, y_n}(\beta_0, \beta_1 \mid y_1, \dots, y_n) \propto \left\{ \frac{1}{S(\beta_0, \beta_1)} \right\}^{n/2}.$$

We will discuss in the next lecture that this is a multivariate t -distribution with $n - 2$ degrees of freedom. It is centered at the least squares estimators of β_0 and β_1 .